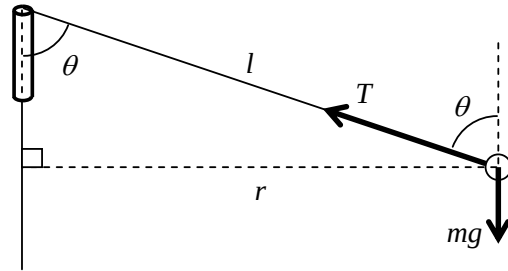


6. Ans: D



For circular motion,

Horizontally, $T \sin \theta = mr\omega^2$ where T = tension in string
 θ = angle between string and vertical.
 m = mass of bob
 r = radius of circular motion.

$$\Rightarrow T \sin \theta = m(l \sin \theta)(2\pi f)^2$$

$$l = \frac{T}{4\pi^2 m f^2}$$

$$l \propto \frac{1}{f^2}$$

Note: T is constant as it is equal to the weight of the brass weights.

7 Ans: B